

MATH 112A - Mathematical Biology

Dani Nguyen

Winter 2025

Contents

1	Review	1
1.1	Ordinary Differential Equations	1

1 Review

1.1 Ordinary Differential Equations

An ODE can be written as $F(y, y', y'', \dots, y^{(k)}) = 0$. The order of the ODE is the order of the highest derivative.

Linear ODEs are a special class of ODEs that have the form:

$$a_0(t)y + a_1(t)y' + \dots + a_k(t)y^{(k)} = f(t)$$

There exists two special cases:

1. If $f(t) = 0$ then the linear ODE is called homogeneous.
2. If $a_i(t)$ are constant and $f(t) = 0$ then we have a linear homogeneous ODE with constant coefficients.

ex

$$y' = 0$$

This is a first order linear homogeneous ODE with constant coefficients.

$$y''' + \sin t \cdot y'' + (t^2 + 1)y + 20t = 0$$

This is a third order linear non-homogeneous ODE

$$y' \cdot y + 20$$

This is a first order non-linear non-homogeneous ODE.

$$y^2 + y'' = 0$$

This is a second order non-linear homogeneous ODE.

The simplest type of ODEs is linear, homogeneous, with constant coefficients.

ex $y'' + y' - 6y = 0$

Look for solutions as:

$$\begin{aligned}y &= e^{pt} \\ y' &= pe^{pt} \\ y'' &= p^2 e^{pt}\end{aligned}$$

$$\begin{aligned}p^2 e^{pt} + pe^{pt} - 6e^{pt} &= 0 \\ p^2 + p - 6 &= 0 \\ p &= \frac{-1 \pm 5}{2} \\ &= -3 \text{ or } 2\end{aligned}$$

The general solution is $y(t) = Ae^{-3t} + Be^{2t}$ Initial value problem: Suppose

$$\begin{aligned}y(0) &= 1 \\ y'(0) &= 0\end{aligned}$$

Then use the general solution to find A and B

$$\begin{aligned}y' &= -3Ae^{-3t} + 2Be^{2t} \\ \begin{cases} y(0) = Ae^{-3 \cdot 0} + Be^{2 \cdot 0} = A + B = 1 \\ y'(0) = -3A + 2B = 0 \end{cases}\end{aligned}$$

Solving the linear system gives $A = \frac{2}{5}$ and $B = \frac{3}{5}$

ex $y' + 2y = e^t$
 $p(t) = 2 \quad q(t) = e^t$
 $E(t) = \exp \int_0^t p(t)dt = e^{2t}$

$$ye^{2t} = e^{3t}$$

$$y = \frac{1}{3}e^t + ce^{-2t}$$

Separable ODEs

$$\frac{dy}{dt} = g(y)h(t)$$

$$\frac{dy}{g(y)} = h(t)dt$$

$$\int \frac{dy}{g(y)} = \int h(t)dt$$

A more realistic model of population growth where r = growth rate and k = carrying capacity.

$$\frac{dy}{dt} = ry(1 - \frac{y}{k})$$

When $y \ll k$, $\text{RHS} \approx ry$